

FUNDAMENTALS OF PROGRAMMING

MOCK FINAL EXAM

Faculty of Information Technology
VNUHCM, University of Science

Time Allotted: 120 Minutes

Problem A: The Symmetrical Array

Time limit: 1.0 second — **Memory limit:** 256 megabytes

You are given an array of n integers. A "Perfect Symmetrical Segment" is a segment of the array that reads the same forward and backward (a palindrome). However, your task is simpler: determine if the entire array is symmetrical and, if so, calculate the sum of all prime numbers within it.

Input

The first line contains an integer n ($1 \leq n \leq 10^3$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^6$).

Output

Print "YES" if the array is symmetrical, followed by the sum of prime numbers in the array. If it is not symmetrical, print "NO".

Example

Input 1:

```
1 5
2 2 3 7 3 2
```

Output 1:

```
1 YES 17
```

Input 2:

```
1 3
2 1 2 4
```

Output 2:

```
1 NO
```

Problem B: String Architect

Time limit: 1.0 second — **Memory limit:** 256 megabytes

You are given a sentence containing lowercase letters and digits. You need to process this string using pointers (dynamic memory).

1. Reverse only the lowercase letters in the string, keeping the digits in their original positions.
2. If a digit is encountered, increase its value by 1 (if the digit is '9', it remains '9').

Input

A single line containing a string S ($1 \leq |S| \leq 1000$).

Output

The modified string.

Example**Input 1:**

```
1 abc12d
```

Output 1:

```
1 dcb23a
```

Input 2:

```
1 hi9
```

Output 2:

```
1 ih9
```

Problem C: Point Management System

Time limit: 2.0 seconds — **Memory limit:** 256 megabytes

You are tracking n points in a 2D plane. Each point is defined by a structure `Point` with `x` and `y` coordinates.

You must:

1. Filter out points that lie in the first quadrant ($x > 0, y > 0$).
2. Calculate the average distance of these filtered points from the origin $(0,0)$.

Input

The first line contains n ($1 \leq n \leq 100$).

The next n lines each contain two integers x_i, y_i .

Output

The average distance is formatted to 2 decimal places. If no points are in the first quadrant, print "0.00".

Example

Input 1:

```
1 3
2 3 4
3 -1 2
4 6 8
```

Output 1:

```
1 7.50
```

Input 2:

```
1 2
2 -1 -1
3 0 5
```

Output 2:

```
1 0.00
```

Problem D: The Linked List Sorter

Time limit: 1.0 second — **Memory limit:** 256 megabytes

You are given a sequence of integers ending with -1. You must store these integers (excluding -1) in a Singly Linked List.

Implement a function to delete all even numbers from the list and then output the remaining odd numbers in the order they were entered.

Input

A sequence of integers a_i ($0 \leq a_i \leq 10^5$), ending with -1.

Output

The modified linked list values are separated by spaces. If the list is empty, print "EMPTY".

Example

Input 1:

```
1 1 2 3 4 5 -1
```

Output 1:

```
1 1 3 5
```

Input 2:

```
1 2 4 6 -1
```

Output 2:

```
1 EMPTY
```

Problem E: Data Persistence

Time limit: 1.0 second — **Memory limit:** 256 megabytes

A file `input.txt` contains n student records. Each record consists of a name (one word) and an integer score.

Your task is to:

1. Read the records from `input.txt`.
2. Find the student with the highest score.
3. Write the name of that student and their score to `output.txt`.

Note: For this problem, assume you are writing the logic that handles these file streams.

Input (`input.txt`)

The first line contains n ($1 \leq n \leq 100$).

The next n lines contain a string *Name* and an integer *Score*.

Output (`output.txt`)

The Name and Score of the top student.

Example

input.txt:

```
1 3
2 Alice 85
3 Bob 92
4 Charlie 78
```

output.txt:

```
1 Bob 92
```

input.txt:

```
1 2
2 John 50
3 Doe 50
```

output.txt:

```
1 John 50
```